

Second-Half Evaluation Report

Clinical-concept extraction of 20 Roman Urdu chief complaints through the offline Qwen2.5 pipeline, scored against gold English references.

Generation model: qwen2.5:latest (local Ollama, temperature 0.0) Generated: 2026-09-20 22:26:00

Method

- Generation: fuzzy Roman Urdu dictionary pass, then Qwen2.5 clinical-concept prompt (temperature 0.0) via the project's offline Ollama pipeline.
- Scoring: symmetrical stop-word removal, underscore/slash token splitting, semantic-equivalence matching (tachycardia<->palpitations, vertigo<->dizziness, etc.), bag overlap.
- Per item: $P = N_{\text{match}} / N_{\text{gen}}$, $R = N_{\text{match}} / N_{\text{ref}}$, $F1 = 2PR / (P + R)$.

Aggregated Statistics - Overall (n=20)

Metric	N	Mean +/- SD

Food-related subset (items 1, 2, 7; n=3)

Metric	N	Mean +/- SD

Key Observations

- F1 across the 20 complaints: min 0.000 (item 7), max 0.833 (item 1), median 0.545.
- Weakest items are 7, 18, 15, 9; their generated concepts recover the fewest gold tokens.
- Food-related subset (items 1, 2, 7): F1 0.460 +/- 0.346.
- Recall is systematically lower than precision: the model names each finding once, while gold references use fuller clinical phrasing ('heart is beating very fast' vs 'palpitations').

Per-Complaint Comparison

SN	Roman Urdu Complaint	Model Concept (Qwen2.5)	Gold Reference	P	R	F1
2	Chest pain left side khana khane ke baad se aur pasina bohat aa raha hai.	chest pain sweating left side	Left-sided chest pain since after eating and excessive sweating.	0.600 0	0.500 0	0.5455
4	Chest pain aur saans phool rahi hai do ghante se, haath thanday ho gaye hain.	chest pain dyspnea cold hands	Chest pain and shortness of breath for two hours, hands have become cold.	0.800 0	0.500 0	0.6154

SN	Roman Urdu Complaint	Model Concept (Qwen2.5)	Gold Reference	P	R	F1
6	Achanak palpitations ek ghante se, known cardiac patient hai.	palpitations	Sudden palpitations for one hour, is a known cardiac patient.	1.000 0	0.200 0	0.3333
8	Mamuli chest mein pressure aadhay ghante se aur ulti jaisa lag raha hai, family mein heart disease hai.	chest pressureodynalgia family history of heart disease	Mild pressure in the chest for half an hour and feeling nauseous, family history of heart disease.	0.800 0	0.400 0	0.5333
10	Shadeed seena jakar raha hai do ghante se.	severe chest pain	Severe chest tightness constriction for two hours.	0.666 7	0.400 0	0.5000
12	Shadeed chest tightness sotay waqt achanak se aur thakan bohat ho rahi hai.	severe chest tightness palpitations	Severe chest tightness suddenly while sleeping and extreme fatigue.	0.750 0	0.428 6	0.5455
14	Mamuli chest mein tez dard achanak se aur bohat kamzori lag rahi hai, family mein heart disease hai.	chest pain palpitations family history of heart disease	Mild chest with sudden sharp pain and feeling very weak, family history of heart disease.	0.833 3	0.500 0	0.6250
16	Halki chest tightness kaam karte waqt se aur pasina bohat aa raha hai, purani cardiac history hai.	chest tightness sweating cardiac history	Mild chest tightness while working and excessive sweating, old cardiac history.	1.000 0	0.571 4	0.7273
18	Tez irregular heartbeat aadhay ghante se aur dard kandhay tak ja raha hai, diabetes ka mareez hai.	palpitations chest pain diabetes	Fast irregular heartbeat for half an hour and pain radiating to the shoulder, diabetic patient.	0.250 0	0.111 1	0.1538
20	Tez seena mein dard aur pasina seedhiyan chadhte waqt se aur dard peeth tak ja raha hai.	severe chest pain sweating radiating to back	Severe chest pain and sweating while climbing stairs and pain radiating to the back.	1.000 0	0.666 7	0.8000